

MMALS EcoSpec 1.0

A Minimal Systemic Architecture for Goal-Conditioned,
Memory-Preserving Adaptive Learning

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Conceptual, philosophical, mathematical, and implementation specification

Abstract

Mycelium-Inspired Mutualistic Adaptive Learning Systems (MMALS) began as a modular continual-learning architecture in which specialist hosts exchange transformed latent signals through an adaptive medium. The programme subsequently developed several partially independent branches: latent geometry, energy-guided routing, functional memory, host specialisation, ecosystem audit, goal-conditioned control, reinforcement-learning meta-routing, and forward-backward universal control. The resulting design space is expressive but risks becoming an accumulation of mechanisms rather than an implementable system.

This paper introduces *MMALS EcoSpec 1.0*, a normative system specification that gives these branches one common state model, one minimal operational loop, one objective interface, and one falsification ladder. Its central design rule is: *use the simplest representation and control mechanism that preserves the information required for correct, stable, economical, and auditable action*. The specification distinguishes observation, latent state, context belief, candidate route, committed action, reconstructive memory, synthetic memory, goal, system health, and resource budget. It defines adaptive-complexity modes that range from compiled single-route execution to sparse multi-hypothesis control and critical high-audit operation. It formalises ecosystem-level contribution, functional route memory, goal-conditioned route energy, compression validity, route compilation, and drift-aware fallback. It then maps the same interfaces to the existing RC2O candidate-trace selector, to a contextual-bandit interpretation, to sequential reinforcement learning, and to forward-backward representations for zero-shot control.

EcoSpec is not an empirical claim that the complete architecture outperforms established continual-learning or reinforcement-learning methods. It is an implementation contract and research protocol. Classical probability-simplex routing is the mandatory baseline. Information-geometric and quantum-inspired density-state representations are optional ablations that must earn their additional complexity under controlled cost and parameter budgets. The accompanying repository contains the LaTeX source, an executable Python reference implementation, configuration and trace schemas, contract tests, a smoke demonstration, and a staged experimental roadmap.

Keywords: continual learning, mixture of experts, adaptive routing, energy-based models, goal-conditioned control, reinforcement learning, functional memory, information geometry, system architecture, auditability.

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1 Introduction

MMALS was initially motivated by a disciplined biological analogy: a mycelial medium should not merely relay signals, but route, transform, reinforce, prune, and preserve exchanges among heterogeneous hosts. Early prototypes therefore moved from static structural routing to a learnable fungal medium, then to delayed mutualism, soft consolidation, output distillation, functional route memory, and route–function disentanglement. The most important internal result of this sequence is conceptual rather than architectural: preserving a route coefficient vector is not equivalent to preserving the function computed through that route. The relevant memory object is mediated computation, not topology alone (MMALS Project, 2026a).

Later branches broadened the programme. Geometry-MMALS asks whether the context–route–host–memory relation forms a stable functional geometry. RC2O provides a transparent candidate-trace selection and continual-learning qualification scaffold. The RL branch treats routing as sequential decision-making under changing goals. Forward–backward control asks whether a reusable representation of reachable futures can support rapid adaptation to new utilities (Touati and Ollivier, 2021; Touati et al., 2022). A quantum-inspired note explores density-state functional memory under ambiguous contexts, while explicitly rejecting any claim of physical quantum advantage (MMALS Project, 2026d).

Each branch is individually understandable. The system-level problem is that they can be assembled in too many ways. A design that contains a context estimator, graph geometry, energy model, multiple memories, a router, a meta-policy, a forward model, a backward goal embedding, host-specific transformations, drift diagnostics, and an audit graph may be intellectually rich yet operationally incoherent. A system cannot be implemented, tested, or falsified unless the role, state, interface, and necessity of every mechanism are explicit.

EcoSpec addresses this problem by replacing an additive architecture with a layered contract:

1. a **mandatory minimal kernel** that can be implemented using the current classical MMALS backbone;
2. **adaptive mechanisms** that activate only under uncertainty, drift, risk, or goal change;
3. **research extensions** that must reduce to the kernel and demonstrate measurable value at controlled cost.

The specification makes six contributions.

1. It states six axioms that connect philosophy, mathematics, and engineering.
2. It defines a complete ecosystem state and a minimal TPUT operational loop.
3. It formalises goal-conditioned energy routing with adaptive complexity.
4. It separates reconstructive and synthetic memory and defines safe compilation.
5. It maps RC2O, RL, and forward–backward control onto one route-decision interface.
6. It provides a claim ladder, baseline matrix, trace contract, and implementation roadmap.

The term *TPUT* is retained as the established MMALS branch label. This paper treats it as the operational control layer and does not rely on a potentially unstable expansion of the acronym. Its precise meaning is given by the state transition and control contracts in this specification.

2 Design Problem and Scientific Position

2.1 From a host ecology to a complete MMALS ecosystem

An MMALS host must not be valued only by isolated predictive accuracy, nor only by its relation to other hosts. Its value is its effect on the complete MMALS ecosystem: context inference, routing, transformations, memory, future retention, stability under drift, resource use, auditability, and the final action. The same principle applies to every element of the system.

Let the ecosystem be denoted by

$$\mathfrak{E} = (\mathcal{O}, \mathcal{Z}, \mathcal{B}, \mathcal{H}, \mathcal{T}, \mathcal{R}, \mathcal{M}^R, \mathcal{M}^S, \mathcal{G}, \mathcal{Q}, \mathcal{L}), \quad (1)$$

where the symbols respectively denote observations, latent representation, belief state, hosts, transformations, candidate routes, reconstructive memory, synthetic memory, goals, health/quality state, and audit ledger. A component $e \in \mathfrak{E}$ is useful if its presence improves system-level utility under a declared evaluation protocol.

A simple counterfactual contribution is

$$\Delta_e = J(\mathfrak{E}) - J(\mathfrak{E} \setminus e), \quad (2)$$

where J is not accuracy alone, but a declared multi-objective system score. Because interactions matter, pair and coalition effects are also relevant:

$$\Delta_{e,f}^{\text{int}} = J(\mathfrak{E}) - J(\mathfrak{E} \setminus \{e, f\}) - \Delta_e - \Delta_f. \quad (3)$$

This prevents the mistaken conclusion that a locally weak component is globally useless. A host may stabilise a route, detect ambiguity, provide a low-cost fallback, or preserve an old function even if its standalone accuracy is mediocre.

2.2 What is preserved and what is allowed to change?

Continual learning exposes a basic tension. A system must preserve useful old behaviour while adapting to new data. Strong isolation can produce zero forgetting by construction but grows capacity or blocks reuse. Strong sharing improves reuse but exposes old functions to interference. Early MMALS prototypes made this tension explicit, and comparative evaluation must include regularisation, replay, expansion, passive mixture-of-experts, and joint-training controls (Kirkpatrick et al., 2017; Zenke et al., 2017; Li and Hoiem, 2018; Rebuffi et al., 2017; Lopez-Paz and Ranzato, 2017; Rolnick et al., 2019; Rusu et al., 2016; Mallya and Lazebnik, 2018; Shazeer et al., 2017).

EcoSpec adopts the following position:

Functional preservation

MMALS should preserve the function needed by old contexts, not necessarily the exact route, host identity, parameter vector, or internal geometry that previously implemented it.

A route may move while function remains stable; a route may remain fixed while function drifts. Therefore, route stability is a diagnostic but not a sufficient memory criterion.

2.3 Compression is not commitment

A common engineering mistake is to equate synthesis with a single label. Let the context posterior be

$$b_t(c) = q_\omega(c \mid x_t, m_t), \quad b_t \in \Delta^{C-1}. \quad (4)$$

Premature commitment chooses

$$\hat{c}_t = \arg \max_c b_t(c). \quad (5)$$

A compact but uncertainty-preserving synthesis instead stores only the top- k support:

$$\tilde{b}_t = \Pi_k(b_t), \quad \|\tilde{b}_t\|_0 \leq k, \quad (6)$$

where Π_k renormalises the largest k entries. This can reduce memory and compute without pretending that an ambiguous case is certain.

The operational rule is:

Compression without premature collapse

Compress early when information is redundant; commit late when remaining uncertainty could change the action, the safety constraint, the memory update, or the audit conclusion.

3 Six Normative Axioms

EcoSpec is governed by six axioms. They are normative engineering rules, not claims about biological mycelium or physical quantum systems.

3.1 Axiom A1: Systemic Contribution

Statement. The value of an element is its contribution to the complete MMALS ecosystem under declared goals, constraints, and counterfactual controls.

Consequence. Host specialisation, router quality, memory quality, and geometry quality are measured by system outcomes and interaction effects, not isolated proxies.

3.2 Axiom A2: Functional Memory

Statement. The primary memory object is the route-conditioned function required for future behaviour.

For host functions g_i , medium transformations T_i , and route weights r_i , define the mediated latent state

$$z_f(x, s) = \sum_{i=1}^H r_i(x, s) T_i(g_i(x)). \quad (7)$$

A functional route record is

$$\Phi(x, s) = \{r_i(x, s), T_i(g_i(x))\}_{i=1}^H, \quad (8)$$

plus sufficient readout information to reconstruct relevant downstream behaviour.

3.3 Axiom A3: Deferred Commitment

Statement. The system preserves a sparse set of relevant hypotheses until an action requires commitment or further uncertainty no longer changes the admissible decision set.

This is not an obligation to execute all hypotheses. It is an obligation not to erase decision-relevant ambiguity solely for implementation convenience.

3.4 Axiom A4: Sufficient Simplicity

Statement. MMALS uses the least complex representation and control mechanism that satisfies current requirements for performance, retention, stability, cost, safety, and auditability.

Let $K(r)$ denote route complexity. The route controller solves a constrained problem rather than maximising raw performance:

$$r_t^* \in \arg \min_{r \in \mathcal{R}_t} C(r) + \lambda_K K(r) \quad (9)$$

$$\begin{aligned} \text{s.t. } & A(r) \geq A_{\min}, R(r) \geq R_{\min}, S(r) \geq S_{\min}, \\ & \text{Risk}(r) \leq \rho_{\max}, \text{Audit}(r) \geq q_{\min}. \end{aligned} \quad (10)$$

This is the architectural form of “the minimum cost that buys defensible confidence.”

3.5 Axiom A5: Goal-Conditioned Control

Statement. Goals parameterise route evaluation and future control; changing a goal should not require rewriting the architecture.

The default goal vector is

$$g_t = (w_A, w_R, w_C, w_D, w_S) \in \Delta^4, \quad (11)$$

corresponding to accuracy, retention, cost, drift stability, and ecosystem specialisation. Additional objectives may be appended if their measurement contract is explicit.

3.6 Axiom A6: Earned Complexity and Falsifiable Reduction

Statement. Every advanced mechanism must reduce to a simpler baseline and must demonstrate incremental value under matched resource and parameter budgets.

Examples include:

- multi-hypothesis routing must reduce to top-1 routing;
- RL control must reduce to one-step energy selection;
- information geometry must reduce to Euclidean/simplex diagnostics;
- density-state memory must reduce to diagonal probabilistic routing;
- compiled memory must fall back to reconstructive execution when its validity envelope fails.

4 Complete Ecosystem State

4.1 State tuple

At decision time t , the operational state is

$$s_t = (x_t, z_t, b_t, \Gamma_t, \mu_t^R, \mu_t^S, g_t, h_t, \beta_t, \ell_t), \quad (12)$$

with:

- $x_t \in \mathcal{X}$: raw or preprocessed observation;
- $z_t = f_\phi(x_t) \in \mathcal{Z}$: shared perceptual representation;
- $b_t \in \Delta^{C-1}$: context belief;
- Γ_t : current route/host/memory graph;
- μ_t^R : reconstructive memory state;
- μ_t^S : synthetic or compiled memory state;
- g_t : goal vector or utility descriptor;
- h_t : system-health vector, including drift and calibration;

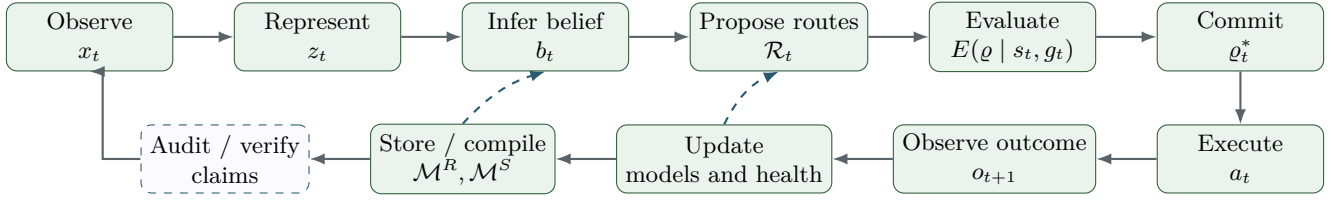


Figure 1: The mandatory TPUT loop. Optional mechanisms change how belief, route proposal, evaluation, and memory are implemented; they do not change the contract between stages.

- β_t : compute, latency, energy, memory, and risk budgets;
- ℓ_t : audit and claim-ledger state.

The state is intentionally broader than an RL environment state. It includes epistemic and governance variables needed to decide how much complexity is justified.

4.2 Route object

A candidate route $\varrho \in \mathcal{R}_t$ is a typed object:

$$\varrho = (\tilde{b}, \mathcal{I}, \mathbf{r}, \mathcal{T}, \mathcal{K}, \mathcal{M}, \mathcal{O}), \quad (13)$$

where:

- $\tilde{b}$ is the context support used by the route;
- $\mathcal{I} \subseteq \{1, \dots, H\}$ is the active host set;
- $\mathbf{r}$ contains host or path allocations;
- $\mathcal{T}$ identifies transformations;
- $\mathcal{K}$ identifies a coalition or execution graph;
- $\mathcal{M}$ declares memory retrieval and update operations;
- $\mathcal{O}$ declares readout and action semantics.

The route is more than a probability vector. It is an executable proposal with predicted cost, risk, memory consequences, and audit depth.

4.3 Typed host interface

Each host H_i exposes

$$H_i : (z_t, \xi_i) \mapsto (u_i, \kappa_i, q_i), \quad (14)$$

where u_i is a representation or proposal, κ_i is execution cost, and q_i is a quality/health report. Hosts may be trainable models, frozen perceptual groves, symbolic solvers, retrieval modules, human-expert channels, or external services. A mature sensory grove is therefore part of the ecosystem even when it is not an adaptive host in the same parameter-update regime.

4.4 System health

A minimal health vector is

$$h_t = (d_z, d_r, d_y, d_T, d_J, \text{ECE}, \text{OOD}, \text{load}, \text{age}), \quad (15)$$

containing latent drift, route drift, output drift, transformation drift, Jacobian/local-sensitivity drift, calibration error, out-of-distribution score, resource load, and memory age. Not every deployment computes every term. The selected health representation itself follows Axiom A4.

5 The Minimal TPUT Operational Loop

The minimal loop in Figure 1 contains ten stages.

1. **Observe**: acquire x_t and metadata.
2. **Represent**: compute z_t using a perceptual adapter.
3. **Infer belief**: update context belief b_t without compulsory top-1 collapse.

Algorithm 1 EcoSpec inference and action**Require:** observation x_t , ecosystem state s_{t-1} , goal g_t , budget β_t

```

1:  $z_t \leftarrow \text{PERCEIVE}(x_t)$ 
2:  $b_t \leftarrow \text{INFERCONTEXT}(z_t, \mu_{t-1}^R, \mu_{t-1}^S)$ 
3:  $m_t \leftarrow \text{SELECTCOMPLEXITYMODE}(b_t, h_{t-1}, \beta_t)$ 
4:  $\mathcal{R}_t \leftarrow \text{PROPOSEROUTES}(z_t, b_t, m_t, \Gamma_{t-1})$ 
5: for all  $\varrho \in \mathcal{R}_t$  do
6:    $(E_\varrho, c_\varrho, v_\varrho) \leftarrow \text{EVALUATEROUTE}(\varrho, s_{t-1}, g_t)$ 
7: end for
8:  $\varrho_t^* \leftarrow \text{COMMITORABSTAIN}(\mathcal{R}_t, E, c, v)$ 
9:  $(y_t, a_t, \tau_t) \leftarrow \text{EXECUTE}(\varrho_t^*, x_t)$ 
10: return  $(y_t, a_t, \tau_t)$ 

```

4. **Propose:** generate a bounded candidate set $\mathcal{R}_t$.
5. **Evaluate:** compute energy, constraints, and predicted future effects.
6. **Commit:** select a route or safe abstention.
7. **Execute:** run the route and produce action/output.
8. **Observe outcome:** collect labels, delayed effects, or environment transitions.
9. **Update:** update permitted parameters, health, and control estimates.
10. **Store/compile/audit:** append reconstructive evidence, update synthetic memory, and verify claims.

The simplest valid implementation may use a frozen encoder, a softmax context estimator, a finite candidate list, a linear energy score, a top-1 route, a supervised loss, and JSON audit records. RL, graph geometry, density matrices, or dynamic topology are not required to satisfy the base specification.

6 Adaptive Complexity: Simple by Default, Rich When Required

6.1 Three operating modes

EcoSpec defines three mandatory modes, each of which may have multiple implementation profiles.

Table 1: Adaptive-complexity modes.

Mode	Entry condition	Representation/control	Memory/audit
Fast	low uncertainty, valid compiled route, low drift/risk	top-1 or narrow top- k ; deterministic energy selection	synthetic memory; compact trace
Ambiguous	competing contexts/routes or moderate drift	sparse belief and route set; optional coalition execution	mixed retrieval; comparative trace
Critical	high risk, OOD, hard retention constraint, regulation, or failed validity envelope	expanded candidates, hard constraints, conservative policy, abstention or human gate	reconstructive memory; full evidence trace

A mode selector can use entropy, energy gap, drift, and risk:

$$u_t = H(b_t), \quad (16)$$

$$\delta_t = E_{(2)} - E_{(1)}, \quad (17)$$

$$\chi_t = \alpha_u u_t + \alpha_d d_t + \alpha_r \text{Risk}_t - \alpha_g \delta_t. \quad (18)$$

Thresholds (θ_F, θ_C) select fast, ambiguous, or critical mode. This rule is deliberately simple; it can later become a learned policy.

6.2 A compression invariance condition

Let $E(\varrho \mid s)$ be the full route energy and $\tilde{E}(\varrho \mid \tilde{s})$ the energy computed from a compressed state. Suppose

$$|E(\varrho \mid s) - \tilde{E}(\varrho \mid \tilde{s})| \leq \epsilon \quad \forall \varrho \in \mathcal{R}. \quad (19)$$

Proposition 1 (ranking preservation). If the energy gap between the best and second-best full-state routes is greater than 2ϵ , compression preserves the selected route.

Proof. Let ϱ_1 and ϱ_2 be the best and second-best full-state routes. Then $E(\varrho_2) - E(\varrho_1) > 2\epsilon$. In the worst case, compression raises ϱ_1 by ϵ and lowers ϱ_2 by ϵ , leaving $\tilde{E}(\varrho_2) > \tilde{E}(\varrho_1)$. $\square$

This proposition supplies a practical compilation test: strong route margin permits aggressive compression; weak margin triggers a richer mode.

6.3 Deferred commitment as action-set dominance

Given context belief b and utility $U(a, c)$, the Bayes action is

$$a_B = \arg \max_{a \in \mathcal{A}} \sum_c b(c) U(a, c). \quad (20)$$

A MAP-context controller selects $\hat{c} = \arg \max_c b(c)$ and then

$$a_M = \arg \max_a U(a, \hat{c}). \quad (21)$$

Because $a_M \in \mathcal{A}$, Bayes optimality gives

$$\sum_c b(c) U(a_B, c) \geq \sum_c b(c) U(a_M, c). \quad (22)$$

The result is elementary but important: preserving the belief does not guarantee perfect decisions, yet it cannot be intrinsically worse than forcing the controller to act as though the MAP context were certain, when both have access to the same action set and utility model.

7 Goal-Conditioned Energy Routing

7.1 Route energy

EcoSpec uses energy as the common comparison language between supervised selection, policy control, and constrained optimisation (LeCun et al., 2006). For route ϱ , state s , and goal g , define

$$E_\theta(\varrho \mid s, g) = w_A \hat{L}_{\text{task}}(\varrho) + w_R \hat{L}_{\text{ret}}(\varrho) + w_C \hat{C}(\varrho) \quad (23)$$

$$+ w_D \hat{D}(\varrho) + w_K \hat{K}(\varrho) + w_Q \hat{Q}_{\text{risk}}(\varrho) \quad (24)$$

$$- w_S \hat{U}_{\text{spec}}(\varrho) - w_M \hat{U}_{\text{mut}}(\varrho) - w_A^{\text{audit}} \hat{U}_{\text{audit}}(\varrho). \quad (25)$$

Terms are normalised to declared ranges before weighting. The five primary goal weights in Equation (11) map to groups of these terms. Hard requirements are not converted into arbitrary soft weights; they remain constraints.

A stochastic router is

$$p_\theta(\varrho \mid s, g) = \frac{\exp(-E_\theta(\varrho \mid s, g)/\tau)}{\sum_{\varrho' \in \mathcal{R}} \exp(-E_\theta(\varrho' \mid s, g)/\tau)}. \quad (26)$$

The fast mode takes $\arg \min E$. Ambiguous mode retains top- k routes. Critical mode may reject all routes if no candidate satisfies hard constraints.

7.2 Mutualistic and systemic contribution

For a route using host i , the local leave-one-out gain is

$$G_i = L(\varrho \setminus i) - L(\varrho). \quad (27)$$

Positive G_i means the host improves the current route. EcoSpec extends this to a system objective:

$$G_i^{\text{sys}} = J_{t:t+T}(\mathfrak{E}) - J_{t:t+T}(\mathfrak{E} \setminus H_i), \quad (28)$$

which can include retention, cost, drift, and future option value. Because exact counterfactuals are expensive, implementation may use periodic ablations, doubly robust estimators, Shapley approximations, or route swaps. The estimator and its uncertainty must be recorded.

7.3 Complexity regularisation

A route complexity term may be

$$K(\varrho) = \alpha_H |\mathcal{I}| + \alpha_E |E_\varrho| + \alpha_M M_{\text{bytes}} + \alpha_L \text{latency} + \alpha_T \text{traceDepth}. \quad (29)$$

The purpose is not to reward small models unconditionally. It is to prevent unearned complexity from masquerading as intelligence.

8 Dual Memory: Reconstructive and Synthetic

8.1 Reconstructive memory

Reconstructive memory $\mathcal{M}^R$ stores the evidence required to reproduce or contest a decision. A normative trace includes:

- observation reference and feature/version hashes;
- context posterior and uncertainty diagnostics;
- candidate routes and rejection reasons;
- goal vector, hard constraints, and budgets;
- route energies and component terms;
- selected route, host coalition, transformations, and model versions;
- outcome, delayed feedback, and memory update;
- claim-verification status and provenance.

Reconstructive memory can be expensive. Its retention policy is risk- and governance-dependent.

8.2 Synthetic memory

Synthetic memory $\mathcal{M}^S$ stores reusable abstractions:

- context and function prototypes;
- compiled macro-routes;
- low-rank route/geometry summaries;
- calibrated energy surrogates;
- policy/value states;
- validity envelopes and fallback pointers.

A compiled route κ is valid only inside an envelope

$$\mathcal{V}_\kappa = (\mathcal{Z}_\kappa, \mathcal{G}_\kappa, \mathcal{H}_\kappa, \mathcal{B}_\kappa, \delta_\kappa, \epsilon_\kappa), \quad (30)$$

covering latent support, allowed goals, health regime, budgets, drift tolerance, and maximum estimated regret.

8.3 Compilation criterion

A candidate macro-route is compiled when

$$n_\kappa \geq n_{\min}, \quad (31)$$

$$\widehat{\text{Regret}}(\kappa) \leq \epsilon_{\max}, \quad (32)$$

$$\widehat{\text{Drift}}(\kappa) \leq \delta_{\max}, \quad (33)$$

$$\text{AuditRecoverability}(\kappa) \geq a_{\min}, \quad (34)$$

$$\text{CostGain}(\kappa) \geq c_{\min}. \quad (35)$$

Compilation is therefore a measured transformation from reconstructive evidence to a cheaper reusable policy, not an irreversible deletion of history.

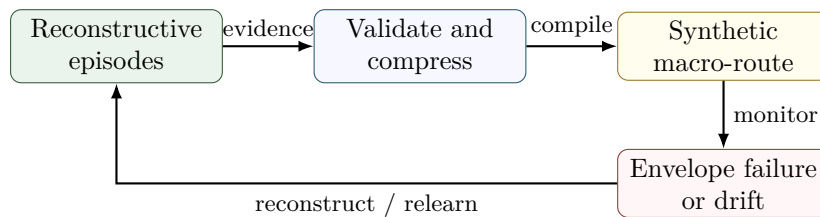


Figure 2: Synthetic memory is a reversible compilation layer over reconstructive evidence.

9 Geometry of the MMALS State

9.1 Mandatory classical geometry

The classical state has a product structure

$$\mathcal{S} = \mathcal{Z} \times \Delta^{C-1} \times \Delta^{R-1} \times \mathcal{M} \times \mathcal{G} \times \mathcal{H}. \quad (36)$$

Euclidean latent distance is not sufficient for every factor. Probability-simplex variables admit divergences and information-geometric metrics (Amari, 2016). A practical block metric is

$$d^2(s, s') = \lambda_z \|z - z'\|_2^2 + \lambda_c d_{\text{JS}}^2(b, b') + \lambda_r d_{\text{JS}}^2(p_r, p'_r) + \lambda_h \|h - h'\|_{W_h}^2. \quad (37)$$

This geometry can support nearest prototypes, drift detection, route continuity, and local trust regions. It does not by itself establish causal specialisation.

9.2 Route graph and transport

Let $\Gamma_t = (V_t, E_t)$ be a typed graph whose vertices may include contexts, hosts, transformations, memory objects, and outputs. Edge attributes encode allocation, cost, confidence, and age. A route is a typed path or small subgraph. Continual update can be interpreted as transport of functional state over Γ_t , constrained by memory and health.

Geometry-G1 questions whether learned routes exhibit functional mediation rather than only nominal clustering. EcoSpec places this question at the diagnostic layer: route continuity, perturbation maps, candidate permutations, route swaps, and held-out contexts determine whether geometry is operationally meaningful.

9.3 Optional density-state extension

For ambiguous contexts, a quantum-inspired extension may represent route-function state using a positive semidefinite trace-one matrix

$$\rho \succeq 0, \quad \text{Tr}(\rho) = 1. \quad (38)$$

Classical routing is the diagonal case

$$\rho_R = \text{diag}(r_1, \dots, r_H). \quad (39)$$

Off-diagonal terms encode learned pairwise coherence or interaction, not physical quantum entanglement. A low-rank implementation can use

$$\rho = \frac{UU^\top}{\text{Tr}(UU^\top)}. \quad (40)$$

The extension is admissible only if it improves performance, memory diagnostics, calibration, or cost-normalised robustness beyond diagonal routing and ordinary covariance/Gram alternatives (Stoudenmire and Schwab, 2016; Schuld and Killoran, 2019; MMALS Project, 2026d). It remains a research branch, not part of the mandatory kernel.

10 Continual Update and Functional Preservation

10.1 Drift decomposition

EcoSpec distinguishes at least five drifts:

$$D_r = \mathbb{E} \|r_{\text{cur}} - r_{\text{ref}}\|_2^2, \quad (41)$$

$$D_z = \mathbb{E} \|z_{f,\text{cur}} - z_{f,\text{ref}}\|_2^2, \quad (42)$$

$$D_y = \mathbb{E} D_{\text{KL}}(p_{\text{ref}} \| p_{\text{cur}}), \quad (43)$$

$$D_T = \mathbb{E} \sum_i \omega_i \|T_{i,\text{cur}}(g_i(x)) - T_{i,\text{ref}}(g_i(x))\|_2^2, \quad (44)$$

$$D_J = \mathbb{E} \sum_i \omega_i \|J_{i,\text{cur}} - J_{i,\text{ref}}\|_F^2. \quad (45)$$

Route-only preservation can yield $D_r \approx 0$ while D_z and D_y remain large. Co-evolution experiments further ask whether D_T or D_J contributes explanatory value beyond D_r, D_z, D_y (MMALS Project, 2026c).

10.2 Update objective

A generic continual update is

$$\mathcal{L}_t = \mathcal{L}_{\text{new}} + \lambda_r D_r + \lambda_z D_z + \lambda_y D_y + \lambda_T D_T + \lambda_J D_J \quad (46)$$

$$+ \lambda_E \mathcal{L}_{\text{energy}} + \lambda_H H(p_r) + \lambda_C C(\varrho) + \lambda_Q \mathcal{L}_{\text{constraints}}. \quad (47)$$

Only terms supported by the active implementation profile are enabled. Coefficients must be validation-selected, and every enabled term requires an ablation.

10.3 Continual update algorithm

Algorithm 2 Continual update with functional memory

Require: new experience batch B_t , reconstructive memory $\mathcal{M}^R$, synthetic memory $\mathcal{M}^S$

- 1: retrieve old-context anchors B_{old} under memory budget
 - 2: compute new-task loss and current route/function states
 - 3: compute enabled drift terms against frozen references
 - 4: apply hard safety and parameter-update masks
 - 5: optimise permitted parameters
 - 6: evaluate all seen contexts using frozen protocol
 - 7: update health, calibration, and claim ledger
 - 8: append reconstructive trace
 - 9: test compilation, invalidation, and fallback conditions
-

11 RC2O Correspondence

RC2O is treated as the current transparent candidate-trace selection and continual-learning qualification backbone. EcoSpec does not replace it. It defines the target interfaces that allow RC2O evidence to become a system-control layer.

Table 2: RC2O-to-EcoSpec correspondence.

RC2O object	EcoSpec interpretation
Candidate trace	Executable route candidate ϱ with features and provenance
Metric evaluator	Measurement map for accuracy, retention, cost, stability, specialisation, and audit
Protocol verifier	Hard constraints, dataset split, seed policy, evaluation timing, and leakage checks
Claim verifier	Claim ledger mapping evidence to allowed, blocked, or provisional claims
Selected candidate	Committed route, policy, or model configuration
Regime detector	Context belief and health state (b_t, h_t)
Eight-dataset scaffold	Offline ecosystem-state and candidate-route benchmark
Goal A–E	Goal vector $g = (w_A, w_R, w_C, w_D, w_S)$
Post-hoc retention evaluation	Delayed transition utility and functional-memory audit
Trace row / CSV artifact	Reconstructive memory record and machine-readable audit event

The RC2O selector can be expressed as a one-step EcoSpec decision:

$$\varrho^* = \arg \min_{\varrho \in \mathcal{R}(s)} E(\varrho \mid s, g). \quad (48)$$

If candidate consequences do not affect future state, this is a contextual bandit. If route selection changes host specialisation, memory, drift, or future candidate availability, the problem is sequential and requires RL or model-predictive control.

11.1 Adapter contract

An RC2O adapter must provide:

1. state extraction from existing metric and diagnostic rows;
2. candidate conversion into typed route objects;

3. goal-conditioned scoring;
4. protocol and claim constraint evaluation;
5. selected-candidate logging in the trace schema;
6. delayed outcome ingestion for RL extension.

The accompanying repository implements a minimal adapter around dictionary/CSV-like candidate rows.

12 Reinforcement-Learning Extension

12.1 MDP and constrained control view

When route decisions have delayed consequences, define an MDP or POMDP

$$(\mathcal{S}, \mathcal{A}, P, R_g, \gamma), \quad (49)$$

where state is the ecosystem state, action is a route/configuration decision, and transition includes learning, memory, host load, and drift. The goal-conditioned reward is

$$R_g(s, a, s') = -E(a \mid s, g) - \lambda_{\text{viol}} V(s, a, s'). \quad (50)$$

A policy $\pi_\psi(a \mid s, g)$ can then learn long-horizon trade-offs (Sutton and Barto, 2018; Haarnoja et al., 2018).

The one-step energy router is recovered when $\gamma = 0$ or when actions have no relevant delayed effect. Thus RL is an extension, not a competing architecture.

12.2 Why route selection is genuinely sequential

Selecting a host changes the data it receives. That changes its future competence. Selecting a synthetic macro-route may reduce audit depth. Repeatedly choosing a dominant host can suppress specialisation elsewhere. Updating a route can protect or damage old functions. Consequently, the controller helps create the ecosystem it later observes.

A minimal transition model is

$$s_{t+1} = F_\eta(s_t, a_t, o_{t+1}) + \varepsilon_t. \quad (51)$$

The RL branch is justified only when a multi-step policy improves declared metrics relative to one-step energy selection under the same candidate set and budget.

12.3 Safe policy hierarchy

EcoSpec recommends a policy hierarchy:

1. hard constraints and safety shield;
2. compiled policy if its validity envelope holds;
3. deterministic energy controller;
4. learned RL policy within the admissible action set;
5. abstention or human gate if no action is admissible.

This ordering prevents a high-capacity policy from bypassing the system contract.

13 Forward–Backward Universal Control Correspondence

Forward–backward (FB) representations learn reusable factors of successor occupancy so that policies for new rewards can be obtained without retraining from scratch (Touati and Ollivier, 2021; Touati et al., 2022). In a simplified notation,

$$M^{\pi_z}(s, a, s^+) \approx F(s, a, z)^\top B(s^+). \quad (52)$$

The backward representation embeds future states or reward support; the forward representation describes route-conditioned reachable occupancy.

EcoSpec maps this structure as follows:

- s : MMALS ecosystem state;
- a : route or coalition decision;
- $F(s, a, z)$: reachable future ecosystem effects under a latent control intent;
- $B(s^+)$: backward embedding of desired future state, constraint satisfaction, or utility;

- z_g : latent control code derived from the current goal vector;
- π_{z_g} : zero-shot or rapidly adapted route policy.

A conceptual FB route score is

$$Q_g(s, a) \approx F(s, a, z_g)^\top z_g, \quad (53)$$

or, with an explicit desired future distribution ν_g ,

$$z_g = \mathbb{E}_{s^+ \sim \nu_g}[B(s^+)]. \quad (54)$$

The controller then selects an admissible action with maximal predicted goal alignment.

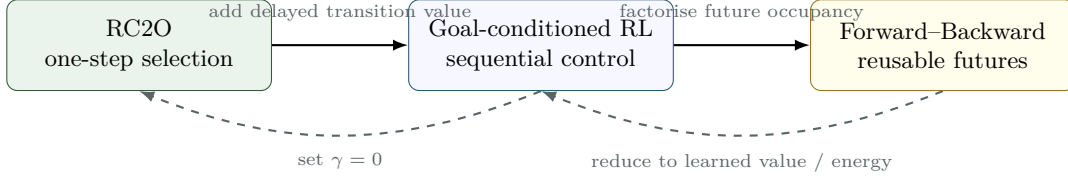


Figure 3: RC2O, RL, and FB are successive control profiles over the same typed route action.

Soft FB generalises the class of utilities and provides a useful future comparison, but it is not a required component of EcoSpec (Bagatella et al., 2026). The first implementation milestone is a transparent sequential meta-router, not zero-shot universal control.

14 Implementation Contract

14.1 Module boundaries

The reference implementation uses the following interfaces:

Table 3: Normative module contracts.

Module	Input	Output
Perceptor	observation	latent vector and quality metadata
Context model	latent, memories, health	context belief and uncertainty
Route proposer	state, mode	bounded candidate route set
Energy model	state, goal, route	energy decomposition and constraint report
Controller	evaluated candidates	committed route or abstention
Executor	route, observation	output/action and execution trace
Memory manager	trace, outcome	reconstructive append, synthetic update, in-validations
Health monitor	state, outcome	drift/calibration/load state
Claim verifier	metrics, protocol	claim status and evidence pointers
RC2O adapter	candidate rows	typed routes and outcomes
RL adapter	transitions	policy/value update

14.2 Reference data classes

The package defines immutable or validated data classes for goals, context beliefs, route candidates, energy breakdowns, trace records, and ecosystem state. The controller must never infer missing constraint semantics from field names. Every metric declares direction, scale, and provenance.

14.3 Feature flags

The default configuration disables research extensions:

- single-step deterministic energy control enabled;
- sparse multi-hypothesis context enabled only above uncertainty threshold;
- synthetic compilation enabled with conservative envelope checks;
- RL disabled;

- forward-backward control disabled;
- information-geometric metric optional;
- density-state memory disabled.

14.4 Trace schema

Every decision trace has a stable identifier and references:

$$\tau_t = (\text{id}, t, x, z, b, g, h, \beta, \mathcal{R}, E, c, \varrho^*, a, o, \Delta\mathcal{M}, \text{claims}). \quad (55)$$

The repository includes a JSON Schema. Production implementations may split sensitive observation references from the audit record while preserving integrity hashes.

15 Experimental Programme and Falsification Ladder

15.1 Stage 0: contract validation

Before model training, test:

- route probabilities and context beliefs are normalised;
- hard constraints cannot be bypassed by energy weights;
- a compiled route invalidates outside its envelope;
- goal changes alter route ranking in a controlled toy system;
- deterministic runs reproduce traces under fixed seeds;
- optional profiles reduce to the classical kernel.

15.2 Stage 1: frozen-backbone route control

Use fixed perceptual features and a frozen host bank. This isolates context, route, goal, memory, and audit behaviour from representation learning. Required demonstrations:

1. route changes under goals A–E;
2. each change improves its target metric or satisfies its constraint;
3. complexity mode changes at declared uncertainty/drift thresholds;
4. synthetic routes fall back correctly.

15.3 Stage 2: RC2O offline trace benchmark

Use the RC2O eight-dataset scaffold as an offline decision dataset. Compare:

- fixed weighted score;
- goal-conditioned linear energy;
- nonlinear energy model;
- contextual bandit;
- oracle selected by future outcomes.

The key metric is not only selected accuracy, but regret against the best admissible candidate under each goal.

15.4 Stage 3: continual-learning backbone

Integrate the selected EcoSpec controller with the current RC2O continual-learning backbone. Mandatory baselines include fine-tuning, EWC, SI, LwF, replay, PNN, sparse MoE, and joint training. Hyperparameters are validation-selected, not arbitrary defaults. Report final average accuracy, forgetting, minimum-task accuracy, backward transfer, memory, latency, training time, parameter count, and GPU memory.

15.5 Stage 4: sequential meta-routing

Construct an environment where current route choice changes future host competence, cost, or retention. Compare deterministic energy selection, contextual bandit, RL, and model-predictive control. RL earns inclusion only if long-horizon utility improves without violating constraints.

15.6 Stage 5: forward–backward control

After the environment and RL transition model are stable, test whether FB representations allow rapid goal switching without per-goal retraining. Compare against goal-conditioned value networks and successor-feature baselines.

15.7 Stage 6: advanced geometry ablations

Only after the classical system is stable, test:

- Fisher/JS geometry versus Euclidean diagnostics;
- covariance/Gram state versus density-state memory;
- real low-rank state versus complex amplitude state;
- diagonal versus off-diagonal route-state models.

An advanced representation is rejected if gains disappear under parameter, memory, and compute matching.

15.8 Claim ladder

Table 4: Claim ladder and minimum evidence.

Claim level	Minimum evidence
C0 – Executable specification	contract tests, deterministic smoke trace, schema validation
C1 – Functional memory	route/function ablation; functional drift aligns with forgetting better than route drift
C2 – Goal-responsive control	route changes across goals and improves goal-specific utility under constraints
C3 – Sequential control value	RL/MPC improves long-horizon utility over one-step energy
C4 – Geometry value	geometric metrics improve prediction/control beyond ordinary baselines
C5 – Density-state value	off-diagonal/low-rank state improves cost-matched ambiguous-context performance or diagnosis
C6 – General MMALS advantage	multi-benchmark, multi-seed, hardened baselines, external replication

16 Relationship to Prior MMALS Versions

Table 5: Research-line mapping into EcoSpec.

Branch/version	Main lesson	EcoSpec location
v0.1–v0.2	structural preservation can reduce interference, but isolation and capacity controls are necessary	route object, complexity cost, baseline and claim ladder
v0.3–v0.4	the medium needs its own cost/utility; immediate mutualistic pressure can destabilise learning	energy decomposition, delayed activation, system contribution
v0.5–v0.7	output and latent preservation matter; specialisation alone is not memory	functional memory axiom and dual-memory update
v0.8–v0.9	separate mechanistic from performance claims; route identity is not function	drift decomposition and claim C1
v0.11	broad comparative harness and cost metrics	stage-3 baseline hardening
v0.12–v0.13	transformation geometry may co-drift; stronger regularisation can over-constrain plasticity	health vector, optional drift terms, earned complexity
v1.x inferred context	explicit task identity must be replaced by latent context belief and guarded routing	state b_t , deferred commitment, complexity modes

Branch/version	Main lesson	EcoSpec location
RC2O	transparent candidate selection, continual qualification, evidence trace	RC2O adapter and one-step controller
Geometry-G1	test functional route mediation, continuity, perturbation, and held-out validity	product geometry and geometry claim C4
TPUT	goal-conditioned operational loop, dual memory, safe control	full EcoSpec kernel
RL/Universal Control	sequential route decisions and goal switching	RL and FB control profiles
QI	mixed route-function memory under ambiguity	optional density-state extension

17 Risks, Non-Claims, and Design Failure Modes

Scientific non-claims

EcoSpec does not claim that MMALS solves continual learning, that the mycelium analogy is causal, that geometry automatically improves control, that density matrices imply quantum advantage, or that the full architecture should be enabled in one implementation.

Major failure modes include:

1. **Architecture inflation:** every research branch is enabled at once.
2. **Proxy capture:** route entropy or host dominance is mistaken for useful specialisation.
3. **Goal laundering:** arbitrary weights produce a desired ranking without validated metric scales.
4. **Audit theatre:** traces record values but not versions, provenance, or rejected candidates.
5. **Compiled-memory lock-in:** cheap macro-routes continue after drift invalidates them.
6. **RL overreach:** sequential control is added where one-step selection is sufficient.
7. **Geometry decoration:** elaborate manifolds do not improve prediction, control, or diagnosis.
8. **Baseline weakness:** poorly tuned EWC, replay, PNN, or MoE artificially inflates MMALS value.
9. **Hidden capacity:** gains arise from more parameters, memory, or compute.
10. **Irreversible ambiguity loss:** early top-1 context selection removes the information needed for safe action.

The correct response to a failed advanced branch is reduction, not rhetorical rescue. The classical kernel remains the control.

18 Discussion

EcoSpec gives MMALS coherence by treating geometry, energy, memory, specialisation, audit, and control as views of one decision process rather than as independent modules. Geometry describes the state and its neighbourhoods. Energy compares admissible routes under a goal. Memory preserves reconstructive evidence and compiles recurrent functions. Specialisation is an ecosystem contribution pattern. Audit records why a route was admissible and why a claim is allowed. RL extends the same route decision across time. Forward-backward representations seek reusable descriptions of reachable futures.

The design is intentionally asymmetric: the simplest classical route is normative; advanced state representations are hypotheses. This asymmetry is necessary. An architecture cannot be “keep it simple” if every conceptual insight becomes a mandatory layer. Simplicity must be encoded in the objective, the mode selector, the reduction tests, and the claim process.

The deepest implication is that abstraction and uncertainty preservation are compatible. MMALS does not need to keep every raw detail in active memory. It can compress observations into latent states, prune beliefs to sparse support, compile repeated route-functions, and reuse macro-policies. What it must preserve is the information capable of changing action, adaptation, or audit. Compression is therefore judged by decision and reconstruction error, not by fidelity to every historical detail.

19 Conclusion

MMALS EcoSpec 1.0 transforms a broad research programme into an implementable system contract. It defines the ecosystem state, the minimal TPUT loop, adaptive complexity, goal-conditioned energy, dual memory, functional preservation, RC2O correspondence, RL extension, and forward–backward control path. It preserves the current MMALS backbone rather than replacing it. It also gives every advanced branch a reduction test and a falsification gate.

The central system principle is:

EcoSpec design rule

Keep the architecture simple; allow behaviour to become complex only when uncertainty, risk, drift, memory, or the goal requires it.

A system that follows this rule can maintain multiple possibilities without carrying unlimited detail, compile repeated experience without losing fallback, and change objectives without retraining its entire architecture. The next scientific step is not another conceptual layer. It is implementation of the classical kernel, contract validation, integration with RC2O traces, and controlled evidence that each additional mechanism earns its cost.

A Notation Summary

Symbol	Meaning
x_t	observation at time t
z_t	perceptual latent state
b_t	context belief distribution
Γ_t	typed ecosystem graph
H_i, g_i	host and host representation function
T_i	medium transformation for host i
ϱ	executable route candidate
$\mathcal{R}_t$	candidate route set
z_f	mediated latent exchange
g_t	goal vector
h_t	system health vector
β_t	resource/risk budget
$E(\varrho \mid s, g)$	goal-conditioned route energy
$\mathcal{M}^R, \mathcal{M}^S$	reconstructive and synthetic memory
D_r, D_z, D_y, D_T, D_J	route, functional latent, output, transformation, and Jacobian drift
ρ	optional positive semidefinite route-function state
F, B	forward and backward representations in FB control

B Normative Pseudocode for Compilation

C Minimal Repository Acceptance Criteria

A release satisfies EcoSpec 1.0 when:

1. the paper compiles without errors;
2. the Python package installs locally;
3. contract tests pass;
4. the toy demonstration shows at least two goals selecting different routes;
5. invalid routes are rejected by hard constraints;
6. a trace validates against the JSON schema;
7. RC2O-like candidate rows can be adapted and scored;
8. optional RL and density-state flags default to disabled;
9. README, version, citation, and licence files are present.

Algorithm 3 Synthetic-memory compilation and invalidation

Require: reconstructive episodes $\mathcal{E}_\kappa$, thresholds $n_{\min}, \epsilon, \delta, a_{\min}, c_{\min}$

- 1: **if** $|\mathcal{E}_\kappa| < n_{\min}$ **then**
- 2: **return** not compiled
- 3: **end if**
- 4: fit candidate macro-route κ
- 5: estimate regret, drift, audit recoverability, and cost gain by held-out episodes
- 6: **if** all thresholds pass **then**
- 7: create validity envelope $\mathcal{V}_\kappa$
- 8: store κ in $\mathcal{M}^S$ with fallback pointers to $\mathcal{M}^R$
- 9: **else**
- 10: retain reconstructive execution only
- 11: **end if**
- 12: **for** each execution of κ **do**
- 13: **if** state outside $\mathcal{V}_\kappa$ or monitored drift exceeds threshold **then**
- 14: invalidate κ and fall back to reconstructive route proposal
- 15: **end if**
- 16: **end for**

Table 7: Illustrative goal profiles; production weights require calibration.

Profile	Accuracy	Retention	Cost	Stability	Specialisation
A: accuracy	0.60	0.15	0.05	0.15	0.05
B: retention	0.20	0.55	0.05	0.15	0.05
C: low cost	0.20	0.10	0.55	0.10	0.05
D: drift stability	0.20	0.20	0.05	0.50	0.05
E: ecosystem specialisation	0.20	0.15	0.05	0.10	0.50

D Example Goal Profiles

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